

Technology #109

Incorporate precisely engineered
microstructures on intraocular lens surface
that manipulate light patterns to promote
natural tear film distribution and meibomian
gland function

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-08-30

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Incorporate precisely engineered microstructures on intraocular lens surface that manipulate light patterns to promote natural tear film distribution and meibomian gland function

WHY DOES IT MATTER?

Documented recurring problem 'adverse_event_without_identified_device_' for intraocular lens: 17 MAUDE reports across 4 distinct devices (20200512 to 20201224); root-cause evidence from 14 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 14 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cus Unresolved gap: Across the 5 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, event, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, producing, recurring, reported, unde

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 4 governing equations, 13 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in optical_photonic. 11-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 2 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 11-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Compare IOLs with different microstructure configurations measuring tear film dynamics, meibomian gland function, and inflammatory marker.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: fluence / irradiance mapping at target depth (tissue phantom). Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 4 equations. Key assumptions: coupling/acceptance angles follow from NA and must be verified on the actual geometry; steady temperature requires a conduction/perfusion model before any quantitative claim; P_{conv} is the converted output power; the conversion fraction $\eta_{\text{conversion}}$ is MODEL_DERIVED until measured; tissue is heterogeneous; μ_a is a MODEL_DERIVED effective value; η under monochromatic NIR differs from solar-rated η ; OPT-003 (Numerical aperture) is applicable only under: paraxial regime; homogeneous medium at the interface.; OPT-004 (Thermal load deposited at target) is applicable only under: steady state; unconverted power deposits as heat.; OPT-001 (Beer-Lambert attenuation) is applicable only under: homogeneous medium; effective coefficient folds scattering into μ_a .; OPT-002 (Photovoltaic conversion) is applicable only under: cell operated below saturation; η at actual wavelength and temperature.; Model input variables requiring sourced values before any numeric result: n (-), θ (rad), I (W/m^2), A (m^2), P_{conv} (W), I_0 (W/m^2), μ_a ($1/\text{m}$), d (m), η (-), I (W/m^2), A (m^2) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: NA (-), Q (W), I (W/m^2), P_{elec} (W) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): source wavelength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; emitted optical power [P_{conv} , W] value_status=UNKNOWN; fiber/transport numerical aperture [NA, -] value_status=UNKNOWN; receiver active area [A , m^2] value_status=UNKNOWN; conversion efficiency [η , -] value_status=UNKNOWN; target-path absorption coefficient [μ_a , $1/\text{m}$] value_status=UNKNOWN; interface temperature rise [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

2 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Passive light manipulation through microstructured surface to enhance tear film stability without active pharmaceutical intervention

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'adverse_event_without_identified_device_' for intraocular lens: 17 MAUDE reports across 4 distinct devices (20200512 to 20201224); root-cause evidence from 14 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 14 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cus Unresolved gap: Across the 5 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, event, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, producing, recurring, reported, unde

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: optical_photonic

Disciplines: optics, photonic engineering, thermal management, electrical engineering

Architecture: PROPOSED architecture (no physical system exists): source (emitter / external illuminator); transport medium (free-space / fiber / tissue); receiver / conversion element; thermal path and safety interlocks — realizing: Incorporate precisely engineered microstructures on intraocular lens surface that manipulate light patterns to promote natural tear film distribution and meibomian gland function

Subsystems:

?: source (emitter / external illuminator) - ?

?: transport medium (free-space / fiber / tissue) - ?

?: receiver / conversion element - ?

?: thermal path and safety interlocks - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Incorporate precisely engineered microstructures on intraocular lens surface that manipulate light patterns to promote natural tear film distribution and meibomian gland function

Key Physics: Optical transport / photonic power governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Optical transport / photonic power governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

OPT-003: Numerical aperture. Expression: $NA = n \cdot \sin(\theta)$. Applicability to this invention: CONDITIONAL; holds under: paraxial regime; homogeneous medium at the interface. Assumptions: coupling/acceptance angles follow from NA and must be verified on the actual geometry. Engaged variables: NOT ESTABLISHED.

OPT-004: Thermal load deposited at target. Expression: $Q = I \cdot A - P_{conv}$. Applicability to this invention: CONDITIONAL; holds under: steady state; unconverted power deposits as heat. Assumptions: steady temperature requires a conduction/perfusion model before any quantitative claim; P_{conv} is the converted output power; the conversion fraction $\eta_{conversion}$ is MODEL_DERIVED until measured. Engaged variables: NOT ESTABLISHED.

OPT-001: Beer-Lambert attenuation. Expression: $I = I_0 \cdot \exp(-\mu_a \cdot d)$. Applicability to this invention: CONDITIONAL; holds under: homogeneous medium; effective coefficient folds scattering into μ_a . Assumptions: tissue is heterogeneous; μ_a is a MODEL_DERIVED effective value. Engaged variables: NOT ESTABLISHED.

OPT-002: Photovoltaic conversion. Expression: $P_{elec} = \eta \cdot I \cdot A$. Applicability to this invention: CONDITIONAL; holds under: cell operated below saturation; η at actual wavelength and temperature. Assumptions: η under monochromatic NIR differs from solar-rated η . Engaged variables: NOT ESTABLISHED.

Assumptions:

- coupling/acceptance angles follow from NA and must be verified on the actual geometry
- steady temperature requires a conduction/perfusion model before any quantitative claim
- P_{conv} is the converted output power; the conversion fraction $\eta_{conversion}$ is MODEL_DERIVED until measured
- tissue is heterogeneous; μ_a is a MODEL_DERIVED effective value
- η under monochromatic NIR differs from solar-rated η
- OPT-003 (Numerical aperture) is applicable only under: paraxial regime; homogeneous medium at the interface.
- OPT-004 (Thermal load deposited at target) is applicable only under: steady state; unconverted power deposits as heat.

- OPT-001 (Beer-Lambert attenuation) is applicable only under: homogeneous medium; effective coefficient folds scattering into μ_a .
- OPT-002 (Photovoltaic conversion) is applicable only under: cell operated below saturation; η at actual wavelength and temperature.
- Model input variables requiring sourced values before any numeric result: n (-), θ (rad), I (W/m^2), A (m^2), P_{conv} (W), I_0 (W/m^2), μ_a ($1/\text{m}$), d (m), η (-), I (W/m^2), A (m^2) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: NA (-), Q (W), I (W/m^2), P_{elec} (W) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): source wavelength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; emitted optical power [P_{conv} , W] value_status=UNKNOWN; fiber/transport numerical aperture [NA , -] value_status=UNKNOWN; receiver active area [A , m^2] value_status=UNKNOWN; conversion efficiency [η , -] value_status=UNKNOWN; target-path absorption coefficient [μ_a , $1/\text{m}$] value_status=UNKNOWN; interface temperature rise [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'adverse event without identified device', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'misalignment loss at couplings', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'fiber bend radiative loss beyond limit', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'photovoltaic conversion degradation with temperature', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'thermal damage at tissue interface', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'window fouling reducing delivered fluence', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Devices matching 'intraocular lens' must maintain function under the conditions of specified occurring injury events with re-	SOURCE_01
DI-002	Mechanism driver (proposed)	Passive light manipulation through microstructured surface to enhance tear film stability without active pharmaceutical in-	MODELLED
DI-003	Expected effect target	Improved tear film stability and reduced inflammation through passive optical modulation without introducing pharmaceutical	MODELLED
DI-004	Domain design input pattern: required applicability NOT ESTABLISHED (candidate pattern; confirm against ENGINEERING PROPOSED	required applicability NOT ESTABLISHED (candidate pattern; confirm against ENGINEERING PROPOSED	ENGINEERING PROPOSED
DI-005	Domain design input pattern: transportability AND ESTABLISHED (candidate pattern; confirm against ENGINEERING PROPOSED	transportability AND ESTABLISHED (candidate pattern; confirm against ENGINEERING PROPOSED	ENGINEERING PROPOSED
DI-006	Domain design input pattern: received applicability NOT ESTABLISHED (candidate pattern; confirm against ENGINEERING PROPOSED	received applicability NOT ESTABLISHED (candidate pattern; confirm against ENGINEERING PROPOSED	ENGINEERING PROPOSED
DI-007	Domain design input pattern: safety extensibility NOT ESTABLISHED (candidate pattern; confirm against ENGINEERING PROPOSED	safety extensibility NOT ESTABLISHED (candidate pattern; confirm against ENGINEERING PROPOSED	ENGINEERING PROPOSED
DI-008	Regulatory design input: compliance standards from IEC 60601-1 (base) and IEC 60601-1-2 (noise), and IEC 60601-1-3 (EMC) is never inferred	compliance standards from IEC 60601-1 (base) and IEC 60601-1-2 (noise), and IEC 60601-1-3 (EMC) is never inferred	UNKNOWN (NEVER INFERRED)
DI-009	Custodied observation: Bilateral Source: https://www.crisis-assessment.com/docs/427445/blood-test-hf-a255e44ce62198647cc314	Source: https://www.crisis-assessment.com/docs/427445/blood-test-hf-a255e44ce62198647cc314	Source: https://www.crisis-assessment.com/docs/427445/blood-test-hf-a255e44ce62198647cc314
DI-010	Custodied observation: Combination of 0.99% cyclosporine and intense blue light (PL) therapy for treatment of severe dry eye disease	Combination of 0.99% cyclosporine and intense blue light (PL) therapy for treatment of severe dry eye disease	Source: https://pubmed.ncbi.nlm.nih.gov/36638603/
DI-011	Custodied observation: A Systematic Review of Safety and Efficacy of Intravitreal Injections in Macular Degeneration	Systematic Review of Safety and Efficacy of Intravitreal Injections in Macular Degeneration	Source: https://www.sciencedirect.com/science/article/pii/S1546222X20300193
DI-012	Custodied observation: Persistent Epithelial Defect After Amblyopia Surgery	Persistent Epithelial Defect After Amblyopia Surgery	Source: https://pubmed.ncbi.nlm.nih.gov/42124746/
DI-013	Custodied observation: Real world topographic surveillance study of AMD and reporting by GPs	Real world topographic surveillance study of AMD and reporting by GPs	Source: https://pubmed.ncbi.nlm.nih.gov/36914329/

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	source (emitter / external illuminator) realizing Incorporate precisely engineered microstructures on circular lens surface that manipulate light patterns	CONCEPTUAL	material choice, circular lens
DO-002	transport medium (free-space / fiber / tissue) realizing Incorporate precisely engineered microstructures on circular lens surface that manipulate light p	CONCEPTUAL	defined microstructure definition
DO-003	receiver / conversion element realizing Incorporate precisely engineered microstructures on circular lens surface that manipulate light patterns to prom	CONCEPTUAL	detailed architecture definition
DO-004	thermal path and safety interlocks realizing Incorporate precisely engineered microstructures on circular lens surface that manipulate light patterns to	CONCEPTUAL	detailed thermal structure definition
DO-005	source (emitter / external illuminator) interfaces with transport medium (free-space / fiber / tissue) via (energy / signal) interface for Incorporate precise	CONCEPTUAL	interface / signal / energy (dimensions, tolerances)
DO-006	transport medium (free-space / fiber / tissue) interfaces with conversion element (free-space / fiber / tissue) via (energy / signal) interface for Incorporate precise	CONCEPTUAL	interface / signal / energy (dimensions, tolerances)
DO-007	receiver / conversion element interfaces with thermal path and safety interlocks: signal / energy / data (dimensions, tolerances) for Incorporate precise	CONCEPTUAL	signal / energy / data (dimensions, tolerances)
DO-008	Device boundaries: patient / fluid inlet and outlet termination / control / data / power is required by selection of precisely engineered microstructures o	CONCEPTUAL	control / data / power is required by selection of precisely engineered microstructures o
DO-009	Flow / force / signal path geometry realizing the mechanism Incorporate precisely engineered microstructures on circular lens surface that manipulate li	UNKNOWN	dimensions, tolerances
DO-010	Operating range for source wavelength within which the mechanism holds its stated behavior / computed bounds for source wavelength; verification of rang	UNKNOWN	stated behavior / computed bounds for source wavelength; verification of rang
DO-011	Operating range for emitted optical power within which the mechanism holds its stated behavior / computed bounds for emitted optical power; verification of rang	UNKNOWN	stated behavior / computed bounds for emitted optical power; verification of rang
DO-012	Operating range for fiber / transport numerical aperture within which the mechanism holds its stated behavior / computed bounds for fiber / transport numerical aperture; verif	UNKNOWN	stated behavior / computed bounds for fiber / transport numerical aperture; verif
DO-013	Operating range for receiver active area within which the mechanism holds its stated behavior / computed bounds for receiver active area; verification of rang	UNKNOWN	stated behavior / computed bounds for receiver active area; verification of rang
DO-014	Operating range for conversion efficiency within which the mechanism holds its stated behavior / computed bounds for conversion efficiency; verification of rang	UNKNOWN	stated behavior / computed bounds for conversion efficiency; verification of rang
DO-015	Operating range for target-path absorption coefficient within which the mechanism holds its stated behavior / computed bounds for target-path absorption coefficient; verifi	UNKNOWN	stated behavior / computed bounds for target-path absorption coefficient; verifi
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism in desired operating position / state / condition / state / condition (sources of data required)	REDESIGN	sensing capability, precisely engineered sensors (sources of data required)
DO-017	Information path: sensing element -> decision / monitor / control / act (minimal); rich data paths are NOT APPLICABLE to this mechanical / physical in	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical / physical in
DO-018	Fixture enabling: fluence / irradiance mapping at target location (issue phantom) - fixture geometry, material, position, selection, knowledge	UNKNOWN	fixture geometry, material, position, selection, knowledge
DO-019	Fixture enabling: conversion efficiency vs wavelength and knowledge - fixture geometry, material, position, selection, knowledge; acceptance instrumentation	UNKNOWN	fixture geometry, material, position, selection, knowledge; acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-020	Fixture enabling: thermal load rise measurement at interfacial region	UNKNOWN — concept name not finalized; no specific instrumentation	Equipment specification; instrument selection; acceptance instrumentation
DO-021	Fixture enabling: tissue-phantom fluence mapping at target depth	UNKNOWN — rig concept finalized; no specific instrumentation	Equipment specification; instrument selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
source wavelength	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
emitted optical power	UNKNOWN (no sourced value)	W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
fiber/transport numerical aperture	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
receiver active area	UNKNOWN (no sourced value)	m^2	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
conversion efficiency	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
target-path absorption coefficient	UNKNOWN (no sourced value)	1/m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
interface temperature rise	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
adverse event without fail	Identified device event without identified device	Design identified device	Device FMEA requires design	UNKNOWN triggered when the operating condition
misalignment loss at coupling	misalignment loss at coupling	couplings: daping tolerance/5/100	5/100 between sensitive fiber and waveguide	the coupled power (Net.O
fiber bend radiative loss	fiber bend radiative loss	bend radius by 6/100	as radius	the bend radius
photovoltaic conversion	photovoltaic conversion	degradation rate at peak/7/damaging output	degradation rate	the degradation rate
thermal damage at tissue	thermal damage at tissue	temperature should be controlled	temperature	the temperature
window fouling reducing efficiency	window fouling reducing efficiency	anti-fouling coating	anti-fouling coating	the anti-fouling coating

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
adverse event without identification	Fails by adverse event without identification	design control NOT ESTABLISHED	design control not established
misalignment loss at couplings	Fails by misalignment loss at couplings	alternator coupling offset between sources	alternator coupling offset between sources
fiber bend radiative loss	Fails by fiber bend radiative loss beyond limit	being below the theoretical bend radius	being below the theoretical bend radius
photovoltaic conversion degradation	Fails by photovoltaic conversion degradation	as a spreading path: a device is positioned at the high voltage contact, the heavily doped region	as a spreading path: a device is positioned at the high voltage contact, the heavily doped region
thermal damage at tissue interface	Fails by thermal damage at tissue interface	interfacial heat transfer concept, the exposed surface of the device is insulated by a thermal barrier	interfacial heat transfer concept, the exposed surface of the device is insulated by a thermal barrier
window fouling reducing delivered fluence	Fails by window fouling reducing delivered fluence	and fouling window coating in a layer that is not a part of the device	and fouling window coating in a layer that is not a part of the device

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Compare IOLs with different microstructures	Compare IOLs with different microstructures	Compare IOLs with different microstructures	Compare IOLs with different microstructures
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test
VF-003	fluence / irradiance mapping at target depth (tissue plane)	fluence / irradiance mapping at target depth (tissue plane)	fluence / irradiance mapping at target depth (tissue plane)	fluence / irradiance mapping at target depth (tissue plane)
VF-004	conversion efficiency vs wavelength and alignment	conversion efficiency vs wavelength and alignment	conversion efficiency vs wavelength and alignment	conversion efficiency vs wavelength and alignment
VF-005	coupled-power monitoring vs alignment	coupled-power monitoring vs alignment	coupled-power monitoring vs alignment	coupled-power monitoring vs alignment
VF-006	bend-loss curve vs radius; output power	bend-loss curve vs radius; output power	bend-loss curve vs radius; output power	bend-loss curve vs radius; output power
VF-007	conversion efficiency vs temperature	conversion efficiency vs temperature	conversion efficiency vs temperature	conversion efficiency vs temperature
VF-008	interface temperature-rise measurement	interface temperature-rise measurement	interface temperature-rise measurement	interface temperature-rise measurement
VF-009	transmitted-power trend through fouled window	transmitted-power trend through fouled window	transmitted-power trend through fouled window	transmitted-power trend through fouled window

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	GaAs / III-V photovoltaic converter	MEM power-over-fiber practice	True	ENGINEERING_PROPOSED
domain candidate material	biocompatible optical window (sapphire/PDMS)	optical fiber/PDMS interfaces	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
	GaAs / III-V photovoltaic converter	1	NOT ESTABLISHED	GaAs / III-V photovoltaic converter	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required
	biocompatible optical window (sapphire/PDMS)	1	NOT ESTABLISHED	biocompatible optical window (sapphire/PDMS)	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- die/wire bonding of converter on carrier: domain registry manufacturing route - candidate only; no process qualification exists
- hermetic optical-window encapsulation: domain registry manufacturing route - candidate only; no process qualification exists
- precision fiber termination and polishing: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'value of critical parameter source wavelength [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter emitted optical power [P_conv W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter fiber/transport numerical aperture [NA -]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter receiver active area [A m^2]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter conversion efficiency [eta -]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter target-path absorption coefficient [mu_a 1/m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'value of critical parameter interface temperature rise [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-007'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}

- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}
- {'unknown': 'acceptance criterion for verification VF-009', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-009'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification	100% compliance	NOT ESTABLISHED	100% compliance	100% compliance
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED	constraint-compliance	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Compare IOLs with diff	NOT ESTABLISHED	VF-001 verification	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment:	NOT ESTABLISHED	VF-002 verification	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	fluence / irradiance mapping at angle	NOT ESTABLISHED	VF-003 verification	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	conversion efficiency	NOT ESTABLISHED	VF-004 verification	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	coupled-power monitor	NOT ESTABLISHED	VF-005 verification	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	bend-loss curve vs rad	NOT ESTABLISHED	VF-006 verification	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	conversion efficiency	NOT ESTABLISHED	VF-007 verification	NOT ESTABLISHED (to be quoted)
WP-10	article for VF-008	per method	interface temperature	NOT ESTABLISHED	VF-008 verification	NOT ESTABLISHED (to be quoted)
WP-11	article for VF-009	per method	transmitted-power trend	NOT ESTABLISHED	VF-009 verification	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 4 governing equations and a 11-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.